DEVELOPING AN INTEGRATED SPECIES DISTRIBUTION MODELLING SYSTEM TO IDENTIFY COMPLEMENTARY CONSERVATION AREAS IN COLOMBIA

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AN INTEGRATED SPECIES DISTRIBUTION MODELING SYSTEM TO IDENTIFY COMPLEMENTARY CONSERVATION AREAS IN COLOMBIA

Knowledge about the geographical distributions of species in present or future scenarios is essential to define surrogates for systematic conservation planning strategies and subsequently prioritize conservation areas that are more likely to support climate change adaptation in conservation actions The Colombia Biodiversity Observation Network (BON) has documented progress in integrating species distribution models (SDMs) with remote sensing information that is relevant for decision- makers via a Bon in a Box named BioModelos (http://biomodelos.humboldt.org.co/). BioModelos is a collaborative online system to map species distributions (Velásquez-Tibata et al, 2019) merging The Alexander Von Humboldt Institute experience running SDMs and a network of experts to be evaluated and refined.

However, distribution maps for most species in the country are too coarse, outdated, and inaccessible to most end users. Currently, the country has web services that allow the integration of species distribution models to produce key information needed to identify new conservation areas. A mapping integrated biodiversity system was needed to create ecological niche models and thus support Colombian agencies' decisions toward identifying complementary conservation strategies in the country.

In this project, we create support tools to integrate, process, and analyze species distribution information with Earth observations (Climate and habitat data). We use this integrated system to establish baseline measurements of the representativeness of biodiversity across multiple regions in Colombia. Our system informs the Colombia Biodiversity Observation Network (BON) via BioTablero and BioModelos, and it helps to plan strategically to define complementary conservation areas resilient to climate change in the country via SIM-SINAP (Monitoring system of the National System of Protected Areas of Colombia).

We ran 8753 models for the same number of species (fishes, plants, mammals, birds, frogs, snakes, insects, and other biological groups, including invasive and introduced species). All the species distribution maps are available online. While the code developed in this project to automate the construction of Species Distribution Models (SDM) from databases gathered, managed and curated is free on github (see below for the tool biomodelos-sdm; https://github.com/PEM-Humboldt/biomodelos-sdm/tree/master).

The models were used to identify complementary conservation areas in Colombia. Our results showed that only 17 Colombian ecoregions of the 67 analyzed have achieved a protection percentage equal to or greater than 30% of areas included within the National System of Protected Areas (SINAP). While approximately half of the ecoregions have protection equal to or greater than 17%. In this scenario, there is a need to identify complementary areas that contribute to the in situ conservation of biodiversity, ecosystem services and cultural values.

The selection and establishment of conservation areas require the use of spatial prioritization protocols that allow maximizing the number of protected species per ecoregion, as well as identifying areas that would serve as connectivity corridors to generate a network of connected protected areas. In this project, area prioritization protocols and connectivity analysis were implemented in present and climate change scenarios. This made it possible to identify corridors that would facilitate the connection between protected areas in the Serranía de San Lucas, the Central Cordillera, the Valleys of the Sinú River, the Alto San Jorge, the Colombian Massif, as well as the Serranía de la Macarena and the foothills of the Cordillera Oriental (http://reporte.humboldt.org.co/biodiversidad/2021/cap3/302/#seccion4). All information is free online through mapping services (https://ee-biomodelos-gee4geo.projects.earthengine.app/view/natgeo).

PRODUCTS

1. **Habitat models for over 4000 species that Biomodels can adapt to support a wide range of management needs.**

The geographic distribution maps were obtained using the “biomodelos-sdm” modeling tool developed within the framework of this project (see below). This system was parameterized to use the machine learning algorithm named Maximum Entropy algorithm (MaxEnt, Phillips & Dudi 2008). In total, eight biological groups were modeled (fishes, plants, mammals, birds, frogs, snakes, insects and other biological groups, including invasive and introduced species), using parameters that reflect all combinations of environmental variables and optimization (Features classes and Beta parameter)(

Muscarella et al. 2014, Radosavljevic & Anderson 2014). All the rasters generated through the modeling phase in the project are accessible at: http://geonetwork.humboldt.org.co/geonetwork/srv/spa/catalog.search#/metadata/0a1a6bdf-3231-4a77-8031-0dc3fa40f21b

The data set contains 8753 raster maps representing documented native and exotica species in Colombia (Figure 1). These maps and others available on BioModelos have been shared with the National System of Natural Parks of Colombia to contribute to the species baseline for SIM-SINAP (Monitoring System of the National System of Protected Areas of Colombia). They will be utilized to calculate representativeness and species richness indicators.

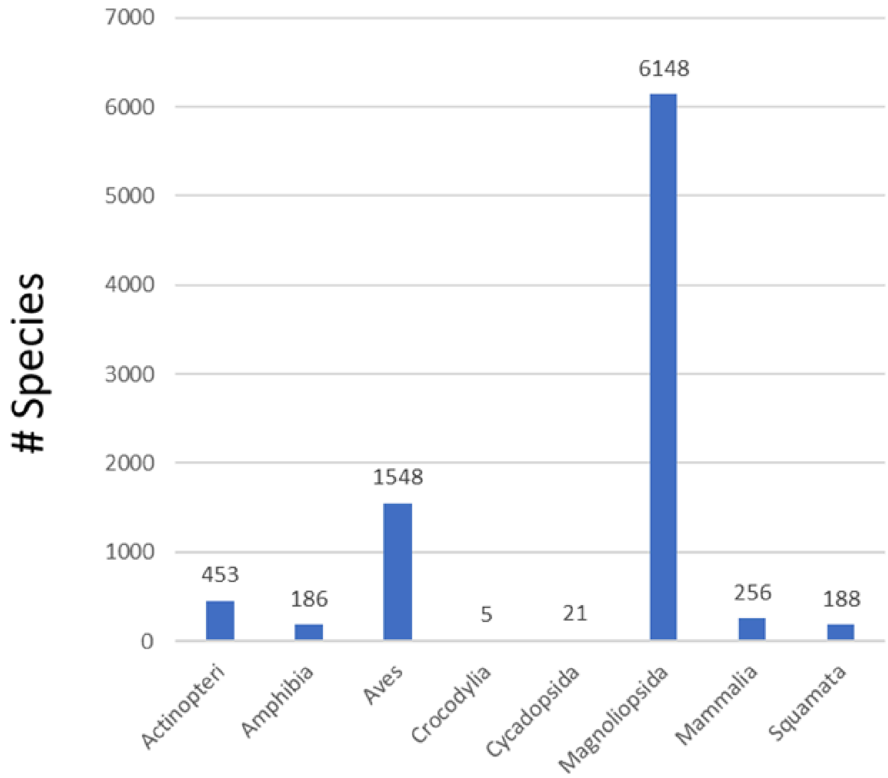


Figure 1. Species grouped by class.

Species records were obtained from GBIF and SpeciesLink, as well as those generated through different projects at the I.Humboldt. The data were processed following standard cleaning procedures such as the elimination of data without coordinates, elimination of geographic duplicates, and validation of the species list with national species lists available on SiBColombia (https://biodiversidad.co/comunidad/proyectos/nacionales/colombiabio/listas/) for example the Colombian Plants Catalog (Bernal et al. 2020). Taking into account that precipitation, temperature and altitude determine the suitability of the species' habitat, 13 Bioclimatic variables (Bio 1, 2, 3, 4, 5, 6, 7, 10, 11, 12, 13, 14, and 15; 30 arc-sen resolution; WGS84 projection system) and altitude were used as predictors in the modeling (Fick & Hijmans 2017).

Ten Percentile was used as a cut-off threshold to select the areas with the greatest suitability for the presence of the species (Radosavljevic & Anderson 2014). Species distribution models (SDMs) frequently project broader geographical ranges than the documented species distribution, encompassing environmentally suitable regions that have yet to be colonized owing to limitations in dispersal. An approach to control overprediction involves integrating spatial information into the fitting process of ecological niche models (ENMs), referred to as a priori methods, or overlapping them with ENM-generated suitability maps, known as a posteriori methods. Therefore the species distribution maps were refined following the Occurrence-based threshold restriction (OBR) approach (Mendes et al. 2020). This approach involves overlaying the binary distribution map without spatial constraints onto species occurrence records and then selecting only the potential distribution areas supported by concrete evidence of the species' presence. This method operates under the assumption that suitable patches that overlap with species occurrences are more likely to be components of species distributions, in contrast to suitable patches that do not intersect with any occurrences (Mendes et al. 2020).

Climate change scenarios can be analyzed at two levels. The first level is based on simulations conducted using General Circulation Models (GCMs), and the second, nested within the GCMs, involves the Representative Concentration Pathways (RCPs), which represent different degrees of greenhouse gas emissions. Therefore, the ecological niche models were projected under climate change scenarios to generate a species distribution baseline for 2030. Relying solely on a single GCM or RCP can lead to prediction errors (Guan et al. 2021), potentially introducing biases into decision-making processes. Four GCMs and two RCPs were selected based on a literature review of their usage in Colombia, considering information from institutions such as IDEAM (Institute of Hydrology, Meteorology, and Environmental Studies) and academia. The review identified 13 documents that describe the use of 1 to 13 GCMs and 1 to four RCPs. Consequently, analyses were conducted with projections up to the year 2030 under two emission scenarios (RCP 1.26, "very strict" emissions path., and RCP 5.85, Emissions continue to increase throughout the 21st century, in the "business as usual" scenario.) using four circulation models (wc2.1_30s_bioc_EC-Earth3-Veg_ssp126_2021-2040; wc2.1_30s_bioc_EC-Earth3-Veg_ssp585_2021-2040;wc2.1_30s_bioc_GFDL-ESM4_ssp126_2021-2040;wc2.1_30s_bioc_MPI-ESM1-2-HR_ssp126_2021-2040;wc2.1_30s_bioc_MPI-ESM1-2-HR_ssp585_2021-2040;wc2.1_30s_bioc_MPI-ESM1-2-LR_ssp126_2021-2040; wc2.1_30s_bioc_MPI-ESM1-2-LR_ssp585_2021-2040). The distribution maps for 2030 scenarios were assembled to obtain single maps per species (Araujo & New 2007, Thuiller et al. 2019).

2. **Maps identifying areas of biodiversity importance critical to preventing extinctions of multiple species in Colombia.**

We identified the priority areas that can be included as complementary conservation strategies considering two main objectives: 1) Maximize the number of protected species by reducing the extinction risk, and 2) Maximize connectivity by including corridors that connect current protected areas. A description of the steps to obtain all the spatial data is presented below. The analyses were carried out in R, version 3.3.1 (R Core Team, 2012) and the workflow (Figure 2) and all the associated codes can be found in the following repositories: https://github.com/afsuarezca/Prioritization_connectivity_Colombia/tree/main and https://github.com/PEM-Humboldt/Prioritization_connectivity_Colombia. All the spatial layers were resampled to 300 m x 300 m in the Magna Colombia SIRGAS projection. The final layers can be accessed and consulted at https://ee-biomodelos-gee4geo.projects.earthengine.app/view/natgeo. The complete maps and dataset are open and free on the institutional repository: http://geonetwork.humboldt.org.co/geonetwork/srv/spa/catalog.search#/metadata/01a9f181-f309-486e-92c3-81c68e49ae30.

Identification of species extinction risk

We used the species distribution models described above to quantify the proportion of the distribution area that is effectively protected for each species within the current network of conservation areas (Butchart et al. 2015, Rodrigues et al. 2004a). After calculating the proportion of the range for each species that overlap with currently protected areas, we performed a “gap analysis” (Rodrigues et al. 2004b) to set a conservation goal for each species based on the concept of “minimum representative area”. The minimum representative area corresponds to the minimum area that should be protected to reduce the extinction risk for each species. It was calculated using a negative exponential function where the species' distribution area that needs to be protected is logarithmically scaled from 10% for species with distributions greater than 250,000 km2, to 100% for species with distributions smaller than 1,000 km2 (Rodrigues et al. 2004a, Rodrigues et al. 2004b). To establish the extra area that should be included within complementary conservation strategies, we subtracted the minimum representative area from the distribution area that is currently protected for each species.

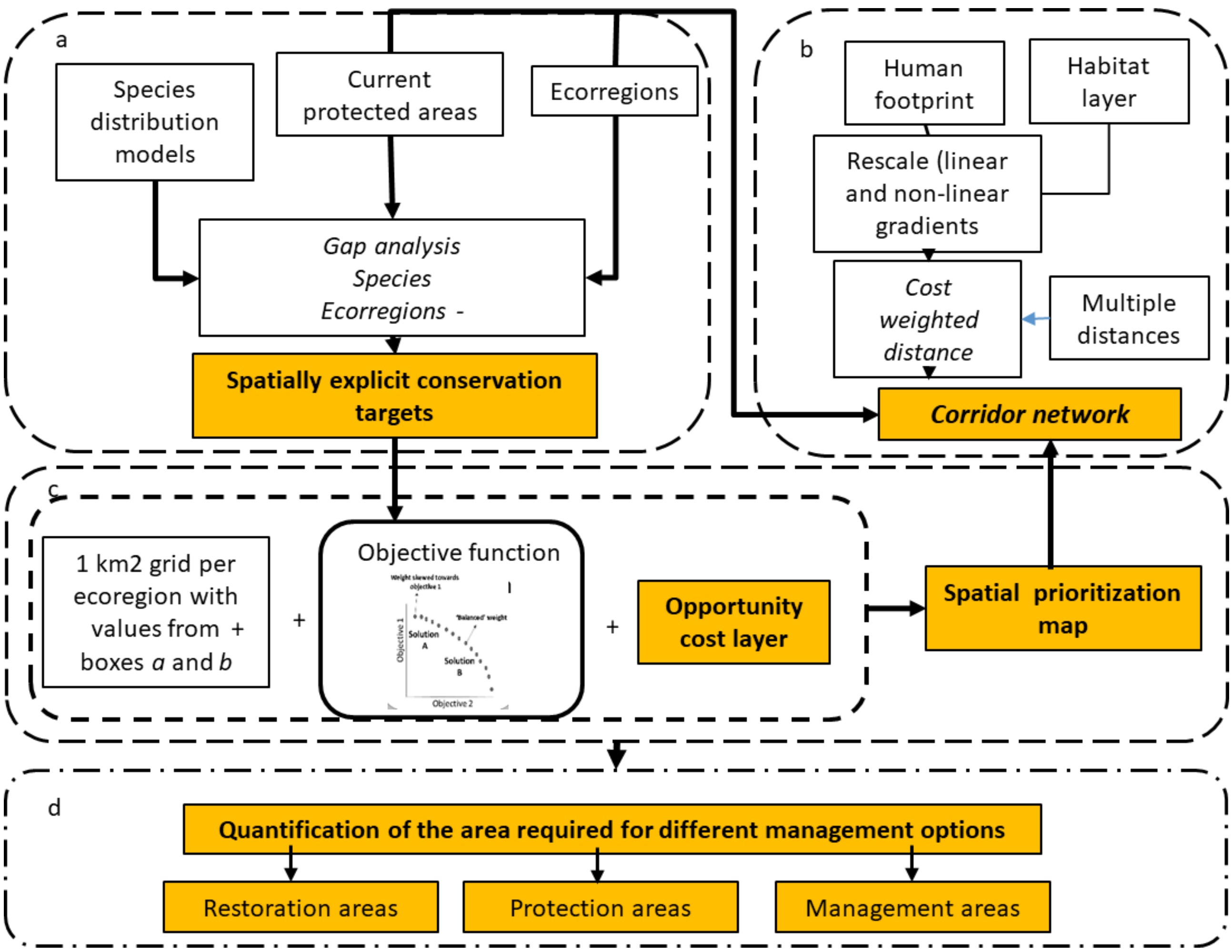


Figure 2. Workflow used to identify areas of biodiversity importance critical to preventing extinctions of multiple species in Colombia. All codes associated with this workflow can be found at https://github.com/afsuarezca/Prioritization_connectivity_Colombia/tree/main. Outputs (in orange) can be found in http://geonetwork.humboldt.org.co/geonetwork/srv/spa/catalog.search#/metadata/211bfe14-f106-4753-9646-

69b42c0a6ba1 The workflow was applied to current species distribution maps and to two climate change scenarios: ssp 126 and 585.

Identification of potential corridors

To identify potential corridors connecting current protected areas, we used Linkage Mapper, a GIS tool designed to develop connectivity analysis (McRae & Kavanagh 2012). Linkage Mapper produces a normalized cost distance surface for a gradient of least-cost paths between core areas (in our case PAs). In this way, it is possible to prioritize polygons based on the mean value of the corridors. Standard Least Cost Paths (NLCCAB) are defined as:

NLCCAB= CWDA+ CWDB−LCPAB

Where CWDA + CWDB corresponds to the cost by adding the cost distance from core areas A and B, and LCPAB corresponds to the values of the lowest cost path. Cells with the lowest values have the greatest potential to function as ecological corridors.

Our analysis assumes that organisms can move more easily between areas with the most intact natural areas according to the human footprint layer (Correa et al. 2020). In order to eliminate the inclusion of unrealistic corridors that could be too long, we limited the maximum geographical distance between two core areas to 200 km, following the protocol described by Belote et al. (2016). Additionally, we limit the width of the corridors to a maximum of 50 km.

Definition of planning units and costs

To define planning units, which consist of potential sites where conservation areas can be selected, the country was divided into squared cells of 1 km2. The planning units only considered terrestrial areas, although they included freshwater bodies if freshwater taxa existed. For each planning unit, we calculated the sum of the contribution to reduce the extinction risk, the mean corridor value, as well as the opportunity costs associated with the transformation of agricultural activities into areas for environmental conservation. The cost layer assumes that producers are motivated to implement strategies to preserve soils when conservation activities produce a monetary benefit equivalent or even higher income from conservation activities than agricultural activities. This is not only intended to protect

the environment from an environmental perspective, but also from a social perspective since conservation strategies become an important source of income for producers. All codes and the workflow used to generate the cost layer can be consulted in the following repository:
http://geonetwork.humboldt.org.co/geonetwork/srv/spa/catalog.search#/metadata/211bfe14-f106-4753-9646-69b42c0a6ba1

We assessed the Net Benefits or profitability obtained in agricultural areas by producers. The Net Profit (NP)(Net farm income) is estimated from the difference between production costs (PC in $COP) and the sales price (SP IN $COP) per unit of production (U) (Eurostat; FAO, 1998; ECLAC, 2005).

$$NP=(SP-PC)*U$$

Production costs (PC) were estimated by mapping main agricultural activities in each planning unit and calculating direct costs (inputs and labor) and indirect costs (land rental, crop management, and crop assistance). This information was obtained from the Price and Supply Information System of the Agricultural Sector-SIPSA (DANE, 2023). In addition, production costs are available in nine specific regions: Antioquia, Cauca-Nariño, Atlantic Coast, Cundiboyacense, Eje Cafetero, Llanos Orientales, Santanderes, Tolima Grande and Valle del Cauca, with information up to 2010 available on the portal. of AGRONET (Ministry of Agriculture, 2010) of the main crops and livestock by region. The value of labor was assumed as the value of the current legal minimum wage.

To map crop production in each planning unit, we obtained information of 312 agricultural species from the National Agricultural Census. These species were grouped into 23 relevant crop classes to optimize cost estimation. For livestock activities, we only considered cattle farming as it corresponds to the main activity that occupies large areas of the territory. According to FEDEGAN data (2015), in Colombia an approximate average of 1.5 to 1.8 heads of cattle per hectare is estimated in extensive production systems. Other farm species, such as pigs and poultry, were not considered since they no longer require large areas of land, and they are generally raised in sheds.

Once crops and livestock were mapped in each planning unit, three main elements were considered to calculate direct and indirect production costs: i) the distance to agricultural supplies distribution centers, ii) the size of the agricultural property, and iii) land rent. Property size was obtained using the information from the National Cadastre (IGAC, 2023). Properties were classified based on size as follows: 1) Microfundio, which covers properties smaller than 3 hectares; 2) Minifundio, which includes properties of 3 to 10 hectares; 3) Small Property, which includes properties of 10 to 20 hectares; 4) Medium Property, which includes properties of 20 to 200 hectares; and 5) Latifundio, which corresponds to properties larger than 200 hectares. Finally, land rent was calculated by classifying the geographical layer of cadastral appraisals (UPRA-IGAC, 2019) into five different categories: 1) areas with an average value less than a current legal minimum wage (SMLV) per hectare, 2) areas with values between 1 and 20 SMLV per hectare, 3) areas with values between 20 and 60 SMLV per hectare, 4) areas with values between 60 and 100 SMLV per hectare, and 5) areas with values greater than 100 SMLV per hectare. All information of production costs was rasterized at 100 m2 (http://geonetwork.humboldt.org.co/geonetwork/srv/spa/catalog.search#/metadata/211bfe14-f106-4753-9646-69b42c0a6ba1).

Identification of complementary conservation strategies

To identify complementary conservation strategies, we used the following objective function:

$$\sum_{i=1}^{I} \frac{\Phi c_i x_i + (1-\Phi) w_i x_i}{Cost_i}$$

Where the parameter Φ represents weights ranging from 0 to 1 and allows the evaluation of trade-offs between objectives (maximizing connectivity and minimizing species extinction risks) by defining their relative importance in each solution (i.e. Φ represents the weight between biodiversity and corridors), ci represents the value of corridors and wi represents the biodiversity value in planning unit i. Cost represents the potential cost of establishing areas for conservation or restoration based on the process described above.

Once the costs (land rent) and benefits (number of species and potential as corridors) are defined for each planning unit, we calculate the optimal solution for a given group of weights. The planning units were ordered from highest to lowest according to cost-benefit values. The units were selected as complementary conservation strategies according to their cost-benefit until a specific target was met (30% protection per ecoregion in the country). This process was applied to the current species distribution models, and repeated for the distribution models developed for two climate change scenarios (ssp 126 and 585), as explained in Product 1.

3. **New/ enhanced software: We developed an integrated Biodiversity Mapping System linking Biomodels and Biotablero Platforms through the development of tool enhancements, services, and connected workflows.**

Modeling system: The "biomodelos-sdm" modeling system is a comprehensive tool designed to automate the construction of Species Distribution Models (SDMs). These functions are built upon a flexible and versatile SDM framework, encompassing various steps and modules (Figure 3). "biomodelos-sdm" is based on the R programming language and its complete source code can be accessed and downloaded from https://github.com/PEM-Humboldt/biomodelos-sdm.

To elucidate its workflow (Figure 3), the system initiates by ingesting input data, including species occurrence records and relevant explanatory variables, while rigorously validating them based on data types and argument selection. Subsequently, the system generates essential files, folders, and a dedicated log file that diligently records events, activities, and pertinent data throughout the execution process. In the following phase, the occurrence data undergoes necessary formatting, and geographic regions for model training and projection are created as user preferences. Notably, a distinctive feature of this system lies in its automatic cropping and masking capabilities for environmental variables under current and future conditions. Furthermore, the system handles occurrence biases and correlations among explanatory variables through advanced analytical tools.

Moving forward, the modeling system proceeds to train SDMs employing one or more algorithms and quantitatively evaluates their performance using a multitude of validation metrics rooted in empirical observations. The subsequent step involves an ensemble approach, where the best models are combined, concurrently calculating measures of variability. Additionally, the system offers an automated solution for users seeking spatial and temporal projections. Finally, upon completing the modeling process, the routine saves the raster surfaces and executes a disk cleanup and space optimization algorithm. This algorithm scans the root folder, identifying and removing redundant files, temporary data, and any superfluous information, thereby reclaiming valuable disk space. This comprehensive feature set ensures the efficient and streamlined operation of the "biomodelos-sdm" modeling system.

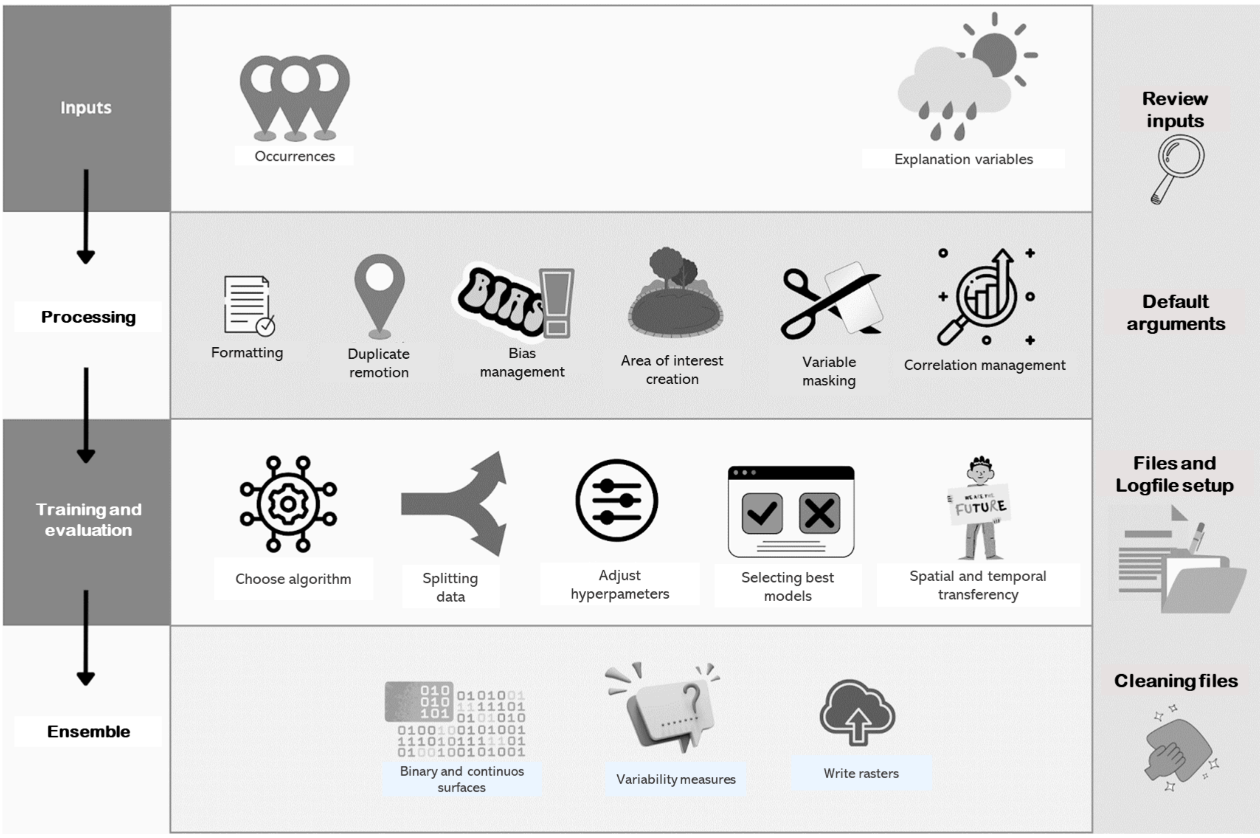


Figure 3. Schematic representation flowchart of the modeling system. An extended version can be found at https://github.com/PEM-Humboldt/biomodelos-sdm/blob/master/modelling/vignettes/flow_data.md

BioModelos platform

Functional enhancements and adjustments

During the years 2022 and 2023, the following updates were made to the BioModelos platform:

- An issue when drawing a polygon was resolved, as in some cases, it was not closing correctly and was being stored incorrectly.
- Adjustments were made to the use cases and associated logic of the species buttons in the viewer.
- Some of the videos in the 'info' section were updated, and the home page received some visual adjustments.
- Translations of some values in the 'Records and Filters' box in the viewer were refined.
- Validation was added for cases where there is no moderator in a group.
- The message displayed for statistical models in the 'Hypothesis' box was updated.

- Some visual adjustments were made related to validations regarding displaying information on the records page and model metadata.

All the changes to the source code are located in the GitHub repository of the BioModelos platform[1].

Public API documentation

BioModelos public API includes a set of 3 endpoints accessible to external users, in contrast to the others, which are exclusively used within the same application. Access to these endpoints is done through a gateway and requires proper credentials. The endpoints are:

- /species/search/{searchWord}: This endpoint allows the user to search for the ID of a specific species. This service allows users to easily and efficiently access the unique ID assigned to each species in our database.
- /species/records/{taxID}: This endpoint provides access to public and curated species records, providing reliable and quality data on the presence of said species, a fundamental first step for generating new distribution models.
- /models: This endpoint is intended to load models generated by experts. This allows researchers and scientists to share and collaborate in developing distribution models, which promotes collaboration and knowledge exchange within the BioModelos community, benefiting the conservation and study of biodiversity.

The documentation of the public API is necessary for the potential users in order to describe how to access and consume the information. OpenAPI is a specification language for HTTP APIs which has been implemented by BioModelos API, using redoc[2] (open source tool for generating documentation), to organize and update the existing documentation that can be found on the GitHub repository[3].

Geoserver adjustments

Geoserver is a tool that serves maps and layers to the BioModelos application as web services of the type WMS (Web Map Service). An important update was implemented to allow BioModelos to serve other users and roles. A new GeoServer user was created, and

[1] BioModelos frontend repository: https://github.com/PEM-Humboldt/biomodelos-website
[2] Redoc documentation: https://github.com/Redocly/redoc
[3] BioModelos API documentation:
https://github.com/PEM-Humboldt/biomodelos-website/blob/develop/public/api_spec_en.yml

a new role and a set of rules were defined to give access from the BioModelos frontend and enable the user to read the current distribution model layers. Also, it was necessary to change the connection parameters from the BioModelos frontend as it is now mandatory to set a user and password to read a layer stored in Geoserver.

Given the above, GeoServer can now manage multiple sets of layers associated with one specific user, his role, and his own access rules.

Tool bm-db-utils

Biomodels bm-db-utils is a set of tools designed to streamline the integration of researchers' work results into the BioModelos platform. The initial utility, developed in 2021, focused on loading and updating raster-format models into the GeoServer used by the Biomodelos platform to display them on the map.

Throughout 2023, additional utilities have been added, primarily related to querying information about ratings and edits made by experts regarding the models assigned to them for evaluation. Additionally, an enhancement was implemented concerning the model loading utility. When a new model is uploaded, the necessary security rules are automatically created to ensure proper access to the GeoServer.

Biomodels bm-db-utils is a command-line client developed in Python. The source code and deployment instructions are in the following GitHub repository: https://github.com/PEM-Humboldt/biomodelos-db-utils.

Uploaded data

Between 2022 and 2023, over 90,000 biodiversity-related data units were added to the BioModelos platform. Occurrence records make up more than 95% of these units as they are the information source for SDM training, encompassing a wide range of taxonomic groups such as epiphytes, insects, armadillos, sloths, anteaters, carnivores, fishes, and more. Additionally, in the same period, approximately 2,300 Species Distribution Models (SDMs) for various taxonomic groups, including endemic birds, rodents, epiphytic plants, bees, beetles associated with pollination, and fishes, were uploaded. These models include probabilistic habitat suitability surfaces and thresholded maps, all developed using the biomodelos-sdm tool. Moreover, SDMs undergoing further refinement through expert mapping, validation, and integration with land cover layers (sensu Velásquez-Tibata et al, 2019) were also incorporated into the BioModelos platform. A comprehensive list of resources published, without restricted units, can be found here https://docs.google.com/spreadsheets/d/13E0ged5OPHYZJoO2ydC2MEAPMM6-4dXgn3cGCCNfUis/edit?usp=sharing.

Website: Biodiversity and climate change

We created a Social Media Campaign to inform interested audiences about the Bes Climate Change website and make it known to them as a valuable resource for accessing crucial information on the relationship between biodiversity, ecosystem services, and climate change. Here, they will find tools and useful information to inspire action and drive positive change.

Main objectives of the campaign

- Announce the Biodiversity and Climate Change website among citizens and those interested in knowing the impacts of climate change and identifying potential sites of special care due to their vulnerability or resilience to the effects of climate change.
- Promote the use of the tool among researchers from public and private companies who identify those areas that could become climate refugees.
- Inform decision-makers and all those interested in developing nature-based alternatives that contribute to the conservation of species and the stability of ecosystems.

Audience

- Citizenship
- Academy
- Researchers
- Public and private entities focused on climate change

Release

Product	**Content**	**Social network**	**Month**
Graphic pieces - small infographics that include link to website	What are climate refugees, and why is it important to identify them?	Instagram LinkedIn	February
Video (screen recording showing the web page) with the narrator talking about what can be found on the web page	Get to know the BES Climatic Change website	Facebook Reel Instagram	February
Instagram or Facebook life	Conversation between Instituto Humboldt and Nat Geo talking about the project	Instagram Facebook	February
Publication #ElDato (Link to website)	Alert figures for biodiversity Figures show the impacts of climate change on the relationship between biodiversity and ecosystem services.	LinkedIn	February
Map (GIF or Image Carousel) showing what can be achieved with the tool	Brief explanation of what can be done with the tool	Facebook Instagram LinkedIn	February

We've developed a web platform that provides training materials (Spanish and English) for the Biomodels-sdm tool, information about biodiversity, conservation, climate change, and the project's products (Figure 4), and more.

Figure 4. The website serves as a central hub for consolidating the results achieved during this project's development (http://proyectos.humboldt.org.co/cambioclimatico/en/home/).

4. **Training resources: We will develop bilingual interactive web-based training, workshops, and webinar materials, which will be openly available and disseminated widely.**

The objective of the training materials is to enhance the capabilities and knowledge of institutions, researchers, and potential users regarding the generation of Species

Distribution Models (SDMs), as well as to provide training on the use of the "biomodelos-sdm" tool. These training materials are designed through practical exercises ("working examples") that delve into the definition of SDMs and how they can be utilized to understand the ecological requirements and habitat preferences of species, identify areas of high species richness, assess the potential impacts of climate change on species distribution, and inform conservation and management strategies. Likewise, the workflow and some of the key features of the "biomodelos-sdm" tool are outlined, along with an explanation and rationale for the methodologies employed by the Institute to automate this process. Finally, questions are presented to stimulate discussions on the strengths and weaknesses of using this information (SDMs) for decision-making. The training materials in English and Spanish language can be found at:

- https://biomodelos.github.io/tallerNatGeo/working-example-1.html
- https://biomodelos.github.io/tallerNatGeo/working-example-2.html
- https://biomodelos.github.io/tallerNatGeo/working-example-3.html
- https://biomodelos.github.io/tallerNatGeo/working-example-4.html
- https://biomodelos.github.io/tallerNatGeo/working-example-5.html

5. **Publications: We anticipate that a number of outputs from the project will be suitable for publication in international peer-reviewed journals. We will produce a paper describing the integrated system, including tool enhancements and workflows and a research paper presenting an analysis identifying complementary conservation areas in Colombia.**

i. Output 1 - *Aquí me quedo: refugios climáticos de las plantas útiles de Colombia en condiciones de cambio climático*, *año 2030* (Climate refugia for useful plants in Colombia under climate change conditions, year 2030). Climate change poses a threat to ecosystems, biodiversity, and communities. Identifying safe regions for resilient plant species is crucial for future conservation and ecosystem services. Potential climate change effects, such as extreme weather events and habitat disruption, impact plant species in Colombia. Colombia hosts a vast array of plant species, including many useful ones. However, some areas are witnessing declines in these species. Species distributions can change due to climate, with some areas remaining stable (climate refugia). Identifying these refuges is essential for species conservation. Using botanical data, future climate refugia were identified for 14 important Colombian plant species. These refugia are mainly in transition zones between the Andes, Amazon, and Pacific regions. Recognizing climate-resilient species is crucial for community adaptation and conservation efforts in the face of climate change (See http://reporte.humboldt.org.co/biodiversidad/2022/cap2/203/).

ii. Output 2 - Listados de plantas potenciales: una respuesta ante la deforestación. To identify flowering plant species impacted by deforestation in Colombia, the distributions of 594 of them, which previously inhabited the so-called deforestation cores, were modeled. The analysis revealed that the selected species have lost values equal to or greater than 14% of their known distribution due to this phenomenon. It is estimated that at least 594 species of flowering plants were affected by deforestation up to 2018. The core regions representing the greatest losses are located in the Amazon (304-367 species), the Pacific (189-367 species), and the Andes in northern Antioquia (321 species) (See. http://reporte.humboldt.org.co/biodiversidad/2022/cap4/403/#seccion3).

iii. Output 3 - *Published:* Áreas complementarias como grandes conectores de la biodiversidad. Over half of Colombia's ecoregions necessitate additional conservation and management areas to reach an internal protection threshold of 30%. By employing protocols prioritizing species preservation and connectivity corridors with these regions, national-level structural connectivity could increase by 15%. Through the implementation of these protocols, corridors have been identified to connect protected areas in the Serranía de San Lucas, the Cordillera Central, the Valleys of the Sinú River, the Alto San Jorge, the Colombian Massif, as well as the Serranía de la Macarena and the foothills of the Eastern Cordillera (See. http://reporte.humboldt.org.co/biodiversidad/2021/cap3/302/#seccion3).

iv. Output 4 - Atlas de la biodiversidad de Colombia: Roedores grandes. This comprehensive resource is designed to serve as a valuable reference for environmental authorities and other stakeholders interested in rodents or animals used as food. http://repository.humboldt.org.co/handle/20.500.11761/36129

v. Output 5 - *In preparation:* Atlas de las Aves Endémicas de Colombia (Atlas of the Endemic Birds of Colombia). We are currently in the process of compiling geographical distributions of endemic bird species for an upcoming publication. This comprehensive resource is designed to serve as a valuable reference for environmental authorities and other stakeholders interested in avian biodiversity.

This publication contains a wealth of information, including detailed species maps, a map depicting species richness, and a comprehensive checklist of potential species found within various administrative areas. We aim to provide a thorough and accessible resource that supports conservation efforts and informed decision-making in avian ecology and habitat preservation.

vi. Output 6 - *In preparation:* An integrated Species Distribution Modeling System to identify complementary conservation areas in Colombia. We are currently in the process of drafting a manuscript that details our innovative species distribution modeling system. This comprehensive document is intended to serve as a valuable resource, complete with repositories and training materials designed to benefit a wide range of stakeholders. With the forthcoming publication, we anticipate a significant expansion in using our modeling system to calculate species distribution models. We aim to make this essential tool accessible to decision-makers and conservationists, fostering widespread and responsible adoption of these models to support informed decision-making and biodiversity conservation efforts.

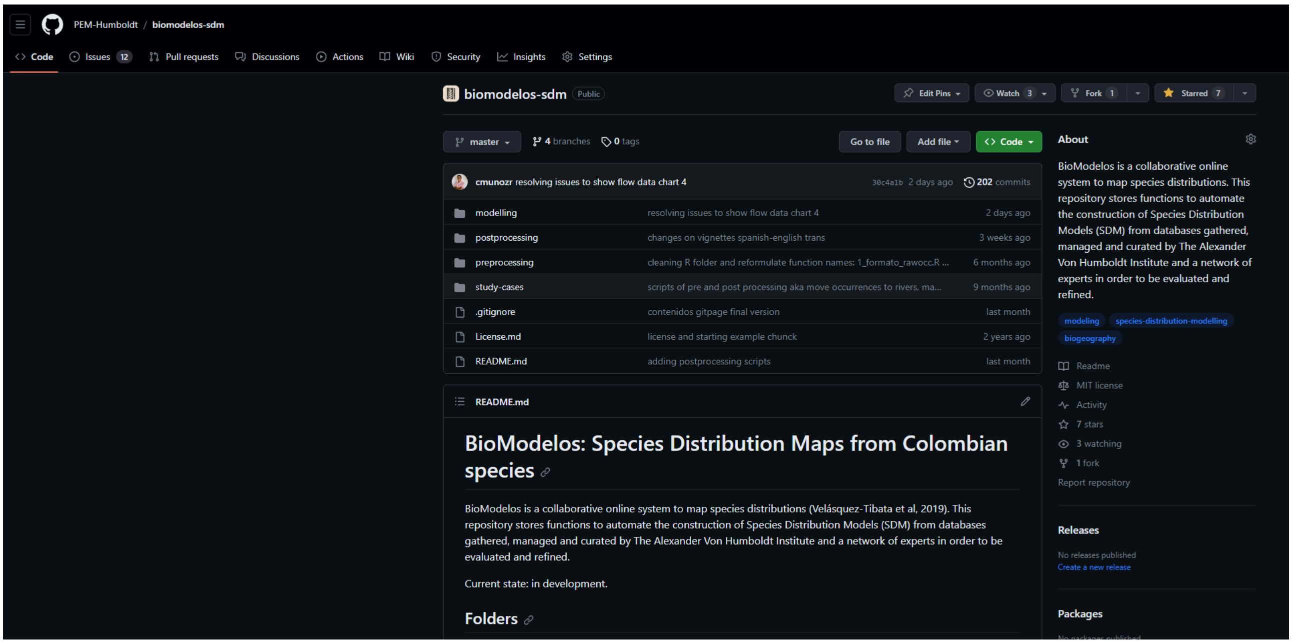
vii. Output 7 - *In preparation: A systematic approach to set spatially explicit complementary conservation areas based on connectivity.* International commitments to formally protect Earth's terrestrial area have prompted calls for far more ambitious targets to preserve biodiversity.. Spatial targets are fundamental to prioritization approaches that guide the efficient allocation of conservation resources and the location of actions (e.g. protection - restoration), while minimizing the costs and negative impacts on resource users. A key target to ensure the long-term maintenance of biodiversity is to promote areas that are more likely to increase connectivity between remaining habitats. However, including connectivity as a conservation feature within multi-objective systematic conservation planning is still challenging. Here, we present a systematic approach to identify complementary conservation areas that increase connectivity among natural areas while accounting for other spatially explicit targets. We used hotspots of richness for different taxa as surrogates for biodiversity. We then compared how connectivity increases once new complementary conservation areas were selected while prioritizing corridors vs other objectives. We hope that the approach presented here will be flexible enough to be applied at regional and local scales. This will facilitate the inclusion of connectivity more explicitly in current prioritization analyses regarding the expected progress on the conservation of areas of particular importance for biodiversity and its contributions to people and the gain in the area, connectivity, and integrity of natural systems

viii. Output 8: - *In revision: A country reaches 30 % of land and ocean conservation area. What's next?* Announcements of countries reaching 30 % of conservation area are widely commended by the international community, despite signs of increasing pressures within protected areas, poor representativity of key threatened ecosystems, and a lack of indicators to monitor the impact of conservation areas on people. Using Colombia's experience, we discuss the need to translate global targets into overarching vegetation retention targets that account for the impacts of conservation actions at local scales on global outcomes. A higher focus is needed on indicators that measure the contribution of conservation areas to the representativeness of ecosystems, how other effective area-based conservation measures (OEMCs) enhance the impacts of the existing protected area network, and how conservation areas promote an equitable and efficient distribution of benefits to people at local, national and global scales. Using multiple examples, we show that reporting the total area under protection without considering this socio-ecological complexity hinders the implementation of informed targets that improve the lives of people who can most effectively manage biodiversity.

INDICATORS USED TO EVALUATE THE WORK

1. **An integrated and comprehensive species distribution modeling system, completed and open-source code available on GitHub**

Introducing biomodelos-sdm, an innovative tool designed to streamline the creation of Species Distribution Models (SDM) using databases that can be collected, managed, and curated by individuals worldwide, even with basic proficiency in the R programming language.

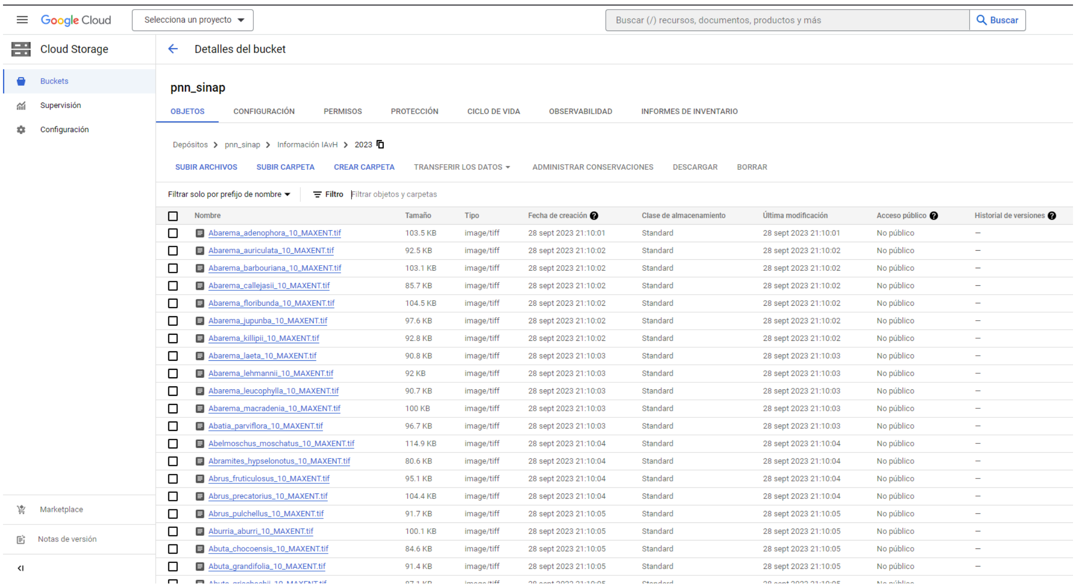


BioModelos: Species Distribution Maps from Colombian species (biomodelos-sdm: https://github.com/PEM-Humboldt/biomodelos-sdm)

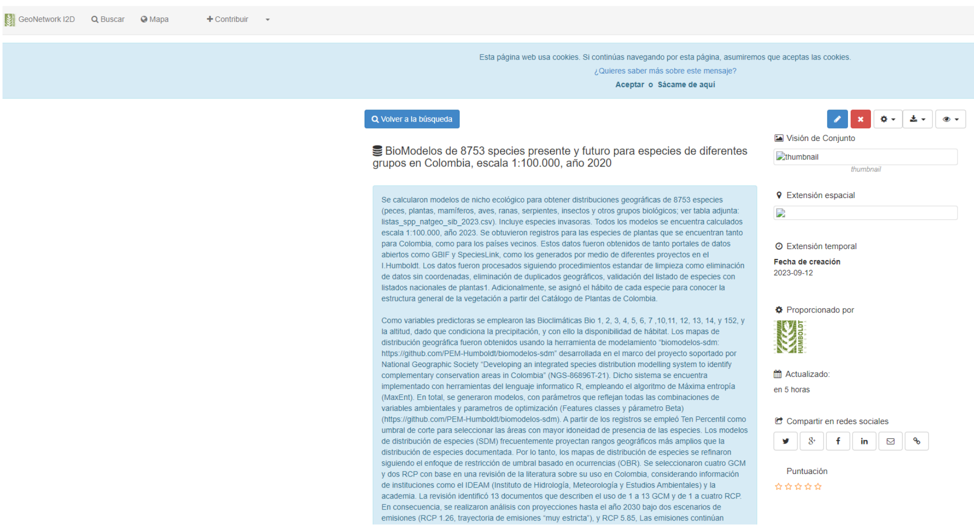
2. **Number of models produced and species processed. We expect to process 4000 species and produce 12000 models representing different climate change scenarios**

We have surpassed the originally proposed model count by an impressive 100%, resulting in 8,753 species modeled and their corresponding versions for various climate change scenarios (total 26,259 models).

These models have been seamlessly integrated into the National Monitoring System of the National System of Natural Parks of Colombia (SIM-SINAP). This integration is crucial in monitoring and supporting Colombia's conservation policy efforts.



SINAP (Sistema Nacional de Parques Naturales de Colombia) Bucket now houses the comprehensive collection of models depicting Colombian species within current scenarios.

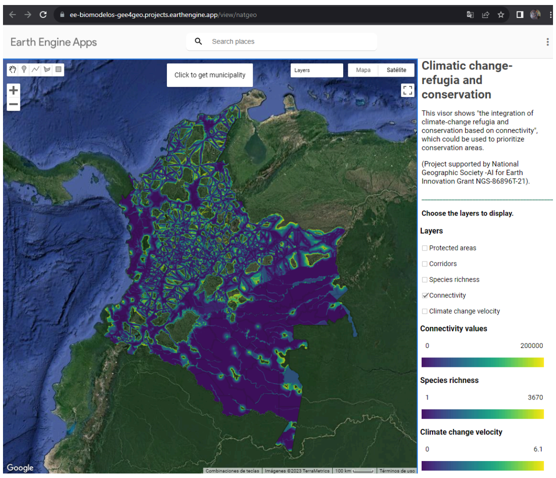


The Institutional Repository hosts models of Colombian species, encompassing both present-day and future scenarios in the context of climate change. http://geonetwork.humboldt.org.co/geonetwork/srv/spa/catalog.search#/metadata/0a1a6bdf-3231-4a77-8031-0dc3fa40f21b

3. **One map showing complementary conservation areas that will serve as climate change refugees in Colombia based on our species distribution models.**

We have created a map of conservation priorities using a framework that considers connectivity and climate change factors. This map has been published and is freely accessible through institutional repositories. Furthermore, in response to user feedback and needs, we have developed a user-friendly application that enables easy access and visualization of these maps. Richness maps were updated with the project results (http://geonetwork.humboldt.org.co/geonetwork/srv/spa/catalog.search#/metadata/92719006-270f-4d17-aba1-69aa42dad92b).

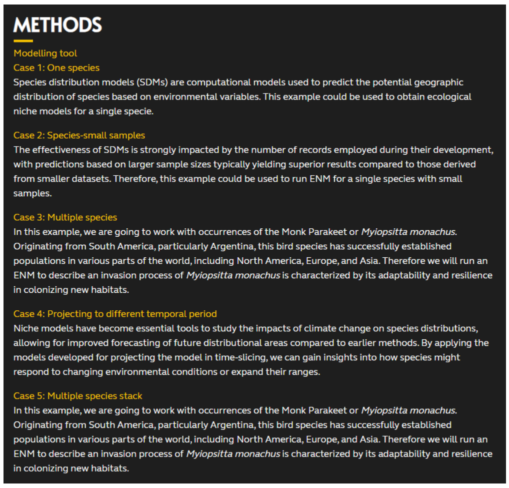
The Institutional Repository hosts a map of complementary conservation areas (http://geonetwork.humboldt.org.co/geonetwork/srv/spa/catalog.search#/metadata/01a9f181-f309-486e-92c3-81c68e49ae30).

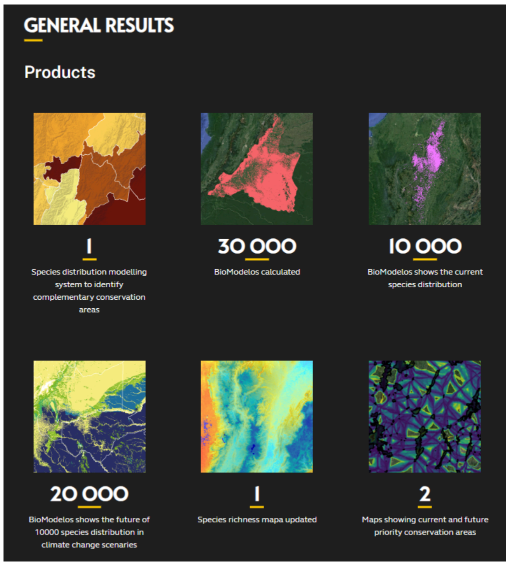


It is possible to access and explore the conservation area map online, alongside other layers of interest (https://ee-biomodelos-gee4geo.projects.earthengine.app/view/natgeo).

4. **Number of participants participating in two different training workshops. During each workshop, we expect 30 participants representing the National Environmental Information System of Colombia, users of the MinAmbiente GUI platform for SINAP.**

We've developed a web platform that not only provides training materials (Spanish and English) for the Biomodels-sdm tool but also enhances our ability to disseminate knowledge about using biodiversity and climate change modeling tools and the project products.





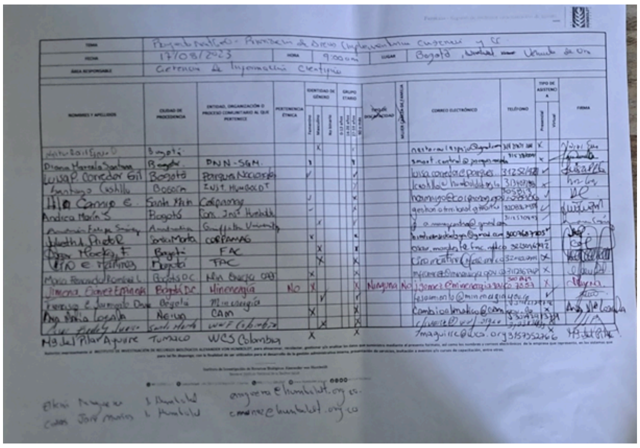
Platform web "Biodiversity and Climatic Change" (http://proyectos.humboldt.org.co/cambioclimatico/en/developing-an-integrated-species-distribution-modelling-system-to-identify-complementary-conservation-areas-in-colombia/)

The Climate Change and Biodiversity web platform was used to support the training workshop contents. As part of our project, we conducted two training workshops as part of our project. The first workshop focused on utilizing the BioModelos-sdm modeling system, while the second aimed to showcase the mapping of complementary conservation areas in Colombia. The workshop materials were sourced from our project repositories, training materials, and the website.

The first workshop attracted the participation of 22 individuals. In contrast, the second had 18 attendees, representing various esteemed institutions, including the National Parks of Colombia, the Aerospace Force of Colombia, the Ministry of Environment and Sustainable Development of Colombia, Regional Autonomous Corporations across the country, the Department of National Planning, Ministry of Mines and Energy, Conservation International, and various non-governmental organizations.

The first workshop (August 17th and 18th, 2023) was attended by people representing the National Parks of Colombia, the Colombian Aerospace Force, the Ministry of Environment and Sustainable Development of Colombia, Regional Autonomous Corporations from across the country, the Department of National Planning, Universities, Conservation International and various non-governmental organizations.

List of Assistants. First workshop (August 24th and 25th, 2023).

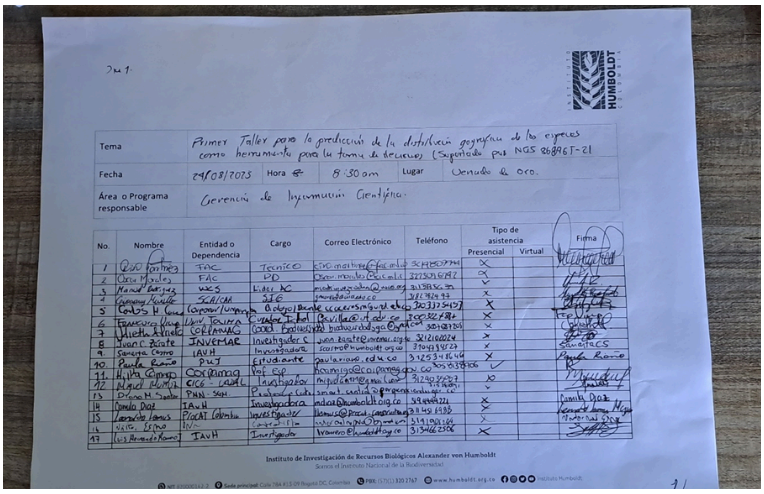


TEMA	Proyecto NatGeo - Presentación de Diseño Implementación Cuenca y CC
FECHA	17/08/2023
HORA	9:00 am
LUGAR	Bogotá, [illegible]
ÁREA RESPONSABLE	Gerencia de Información Científica

NOMBRES Y APELLIDOS	CIUDAD DE PROCEDENCIA	ENTIDAD, ORGANIZACIÓN O PROCESO COMUNITARIO AL QUE PERTENECE
[illegible]	Bogotá	
Diana Marcela Santana	Bogotá	PNN - SGM
Luisa F. Corredor Gil	Bogotá	Parques Nacionales
Santiago Castillo	Bogotá	Inst. Humboldt
[illegible]	Santa Marta	Corpamag
Andrea Marín S.	Bogotá	Cons. Inst. Humboldt
[illegible]	[illegible]	[illegible]
[illegible]	Santa Marta	CORPAMAG
[illegible]	Bogotá	FAC
[illegible]	Bogotá	FAC
[illegible]	Bogotá DC	Min Energía [illegible]
Jimena Gómez [illegible]	Bogotá DC	Minenergía
[illegible]	Bogotá	Minenergía
Ana María Lozada	Neiva	CAM
[illegible]	Santa Marta	WWF Colombia
Ma del Pilar Aguirre	Tumaco	WCS Colombia

The second workshop (August 24th and 25th, 2023), attended by people representing the National Parks of Colombia, the Colombian Aerospace Force, the Ministry of Environment and Sustainable Development of Colombia, Regional Autonomous Corporations from across the country, The José Benito Vives de Andréis Institute of Marine and Coastal Research, Department of National Planning, Universities, Conservation International and various non-governmental organizations

List of Assistants. Second workshop (August 24th and 25th, 2023).



Dia 1.

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Tema	Primer Taller para la predicción de la distribución geográfica de las especies como herramienta para la toma de decisiones (Soportado por NGS 86896T-21
Fecha	24/08/2025 · Hora: 8:30 am · Lugar: Venado de oro.
Área o Programa responsable	Gerencia de Información Científica.

No.	Nombre	Entidad o Dependencia	Cargo	Correo Electrónico	Teléfono	Tipo de asistencia: Presencial	Tipo de asistencia: Virtual	Firma
1	Ciro Martínez	FAC	Técnico	ciro.martinez@fac.mil.co	3147807744	X		[illegible]
2	Oscar Morales	FAC	PD	oscar.morales@fac.mil.co	3273096797	X		[illegible]
3	Manuel Rodríguez	WCS	Líder AC	mrodriguez@wcs.org	3115835630	✓		[illegible]
4	Giovanny Murillo	SCA/CAR	SIG	[illegible]	3182782497	X		[illegible]
5	Carlos H. [illegible]	Corporar/Unipamplona	Biólogo/Docente	[illegible]	3203375457	X		[illegible]
6	Francisco Villa	Univ. Tolima	Curador Ictiol.	fcvilla@ut.edu.co	300322[illegible]	X		[illegible]
7	Ulietta Alnate	CORPAMAG	Coord. Biodiversidad	biodiversidad[illegible]@gmail.com	3007087205	X		[illegible]
8	Juan C. Zárate	INVEMAR	Investigador C	juan.zarate@invemar.org.co	3212102024	X		[illegible]
9	Samanta Castro	IAvH	Investigadora	scastro@humboldt.org.co	3104794527	X		SamantaCS
10	Paula Riaño	PUJ	Estudiante	paulariano.edu.co	3125348644	X		Paula Riaño
11	Hilda Camargo	Corpamag	Prof. Esp.	hcamargo@corpamag.gov.co	3058138906	✓		[illegible]
12	Miguel Muñoz	CIGG - CAL	Investigador	miguelam@gmail.com	3124035987	X		[illegible]
13	Diana M. Santos	PNN - SGM.	Profesional Cont.	smart.santa@[illegible]	[illegible]	✓		[illegible]
14	Camila Díaz	IAvH	Investigadora	mdiaz@humboldt.org.co	3194444221	X		Camila Díaz
15	Leonardo Lemus	PlocAT Colombia	Investigador	llemus@procat-conservation.org	3114516988	X		[illegible]
16	Victoria Espino	[illegible]	Contratista	[illegible]	3141901164	X		Victoria [illegible]
17	Luis Hernando Romero	IAvH	Investigador	lromero@humboldt.org.co	3134662506	X		[illegible]

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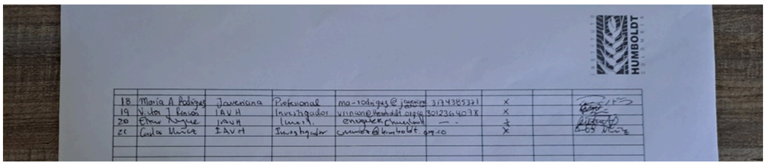

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No.	Nombre	Entidad o Dependencia	Cargo	Correo Electrónico	Teléfono	Presencial	Virtual	Firma
18	María A. Rodríguez	Javeriana	Profesional	ma-rodriguez@javeriana.edu.co	3174385371	X		[illegible]
19	Víctor J. Rincón	IAvH	Investigador	vrincon@humboldt.org.co	3012364078	X		[illegible]
20	Elkin Noguera	IAvH	Invest.	enoguera@humboldt	—	X		[illegible]
21	Carlos Muñoz	IAvH	Investigador	cmunoz@humboldt.org.co		X		Carlos Muñoz

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